

■ 1.4.3 Assigning Symbolic Values

Just as you can assign numerical values to variables in *Mathematica*, as discussed in Section 1.2.2, so also you can assign symbolic values.

This assigns *y* to have the symbolic value *y*. *In*[1]:= *y* = 1 + *x*
Out[1]= 1 + *x*

Whenever you ask for *y*, you now get the expression 1 + *x*. *In*[2]:= *y*
Out[2]= 1 + *x*

When you type in an expression involving *y*, *Mathematica* replaces *y* everywhere by the value you assigned to it. *In*[3]:= 1 - *y*²
Out[3]= 1 - (1 + *x*)²

This clears the value you assigned to *y*. *In*[4]:= *y* =.

Now *y* simply “stands for itself”, and is no longer replaced by 1 + *x*. *In*[5]:= 1 - *y*²
Out[5]= 1 - *y*²

In most computer languages, variables are essentially used just as names to refer to objects like numbers that are generated at intermediate stages in computations. In *Mathematica*, as in standard mathematics, you can also use variables like *y* in a purely symbolic fashion.

If you assign no value to *y*, then any time *y* appears in an expression, *Mathematica* will leave it unchanged. But even though *Mathematica* does not change *y* itself, it can still apply transformation rules to expressions that involve *y*. So, for example, it will replace *y*-*y* by 0 even though you have given no specific value to *y*. The point is that rules like this are correct whatever the value of *y* may be, so they can be used even when *Mathematica* knows no specific value for *y*.

When you use *Mathematica*, you will usually have some variables with specific values, and some which “stand for themselves” and have no value defined. It is important not to confuse which variables you are using in which way at a particular point in your *Mathematica* session.

One of the most common mistakes in using *Mathematica* is to assign a value to a variable at one point, and then to forget this, and try to use the variable in a purely symbolic fashion much later. To avoid this problem, it is important that you remove values for variables when you have finished using them. You can always check whether a particular variable has a value at a particular time just by typing

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the variable, and seeing what result you get.

Always remove values for symbols using $x = .$ when you have finished with them

A very important point to remember when using symbols in *Mathematica*.